

Customer Behavior Analysis

Project Overview

In this project, we analyzed **customer shopping behavior using transactional data from 3,900 purchases collected over a period of time**. The main objective of the project is to understand customer spending patterns, identify which types of customers contribute the most to sales, analyze customer behavior across different age groups and categories, and generate actionable insights that can help the business improve its future sales strategy.

Key Points Covered

1. What type of data are we studying?

We are working with **customer transactional and behavioral data**. The dataset contains information related to customer demographics, purchase details, product categories, purchase frequency, spending patterns, discounts, reviews, and other customer-related attributes.

This allows us to analyze **who the customers are, what they purchase, how frequently they purchase, and how much they spend**.

2. Why are we analyzing this data?

The purpose of the analysis is to understand customer behavior and identify patterns that can support better business decisions.

Through this analysis, we aim to:

- Identify high-value customer segments.
- Understand spending behavior across different age groups.
- Identify the most and least popular product categories.
- Analyze purchasing frequency and customer preferences.
- Understand the impact of discounts and other factors on purchasing behavior.
- Identify opportunities to improve customer engagement and retention.
- Provide data-driven recommendations for future sales and marketing strategies.

3. How are we going to analyze the data?

We will follow a structured data analysis process using **Jupyter Notebook, SQL/ PostgreSQL, and**

Power BI. Step 1: Data Loading, Cleaning & Formatting – Jupyter Notebook

We will first load the dataset into **Jupyter Notebook using Python**. We will inspect the dataset and perform data cleaning and preprocessing.

This will include:

- Checking the structure and data types.
- Identifying missing values.
- Removing or handling duplicate records.
- Correcting inconsistent data.
- Formatting columns and dates.
- Creating new calculated columns where required.
- Performing initial exploratory data analysis.

Step 2: Detailed Analysis – SQL/ PostgreSQL

After cleaning and preparing the data, we will connect the dataset with **PostgreSQL** and use SQL queries to perform detailed analysis.

SQL will help us calculate and summarize important business metrics such as:

- Total revenue and sales.
- Average purchase amount.
- Sales by product category.
- Customer spending by age group.
- Purchase frequency.
- Top-performing products/categories.
- Customer segmentation.
- Other KPIs required for business decision-making.

This step will help us convert the raw data into **meaningful and summarized insights**.

Step 3: Visualization & Reporting – Power BI

Once the SQL analysis is completed, we will connect the processed data to **Power BI** and create an interactive dashboard.

The dashboard will present important KPIs and visualizations in a simple and understandable way so that **team members, managers, or non-technical stakeholders can understand the business insights without going through the raw data**.

The dashboard may include:

- Total Customers
- Total Revenue
- Average Purchase Amount
- Average Review Rating
- Sales by Category
- Sales by Age Group
- Customer Segmentation
- Purchase Frequency
- Top Products/Categories
- Discount/Subscription-related insights

4. Final Business Recommendations

The final stage of the project will focus on converting the analysis into **actionable business recommendations**.

Based on the insights identified from Python, SQL, and Power BI, we will prepare recommendations around:

- Which customer segments should be targeted.
- Which product categories require more focus.
- How to improve customer retention.
- Where discounts can be used effectively.
- Which customer groups have higher spending potential.
- How future marketing and sales strategies can be optimized.

The ultimate goal is not just to create a dashboard, but to **use data to understand customer behavior and support better business decisions and future growth strategies**.

1. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discontinued
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207569	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

2. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:0

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.
9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

3. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



4. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.